

[RI/2025/Q4]

4 Your school limits assembly announcements to 300 seconds per day. Announcements that cannot fit in 300 seconds are deferred to the next day.

Announcements are read in order of priority: High → Medium → Low. Three queues are used for each of these priorities.

Announcements are stored in two files:

Both of these files follow the same format:

<department>,<priority>,<duration>,<title>

e.g. Discipline,High,90,Uniform Violation Rules

Implement the Announcement Scheduler, using provided code template, scheduler.py, which

- Load announcements from announcement.txt and deferred_announcement.txt
- Schedule announcements up to 300 seconds/day: High → Medium → Low priority (FIFO within priority)
- Display today’s schedule, clear announcement.txt, and archive to read_announcements_archived.txt
- Save leftover announcements to deferred_announcement.txt

Rename scheduler.py as

Task4_<name>_<index number>.py

[RI/2025/Q4 Cont ']

Task 4.1

Complete the `Announcement` class, which stores these attributes:

- `department`
- `priority`
- `duration`
- `title`

Implement the following methods:

- `__str__()`: Return a human-readable string representation of the announcement. Example: `"CCA - Science Fair (90s)"`
- `csv_str()`: Returns a csv-formatted string for writing to file. Example: `"CCA,High,90,Science Fair"`

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Task 4.2

Complete `Queue` Class that stores `Announcement` objects, with suitable constructor and attribute(s).

You should implement these methods:

- `enqueue()`: Adds an announcement to the queue
- `dequeue()`: Remove and return the item at front of the queue, if not empty
- `is_empty()`: Returns `True` if queue is empty, `False`, if otherwise
- `output(file)`: Write each announcement in the queue to provided file, one per line, using `Announcement.csv_str()` method

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Task 4.3

The `AnnouncementScheduler` class serves as the central controller of the entire announcement scheduling process :

AnnouncementScheduler		
Attributes		
Identifier	Data Type	Description
<code>time_limit</code>	<code>int</code>	Default of 300
<code>deferred_filename</code>	<code>str</code>	<code>deferred_announcement.txt</code>
<code>announcement_filename</code>	<code>str</code>	<code>announcement.txt</code>
<code>archive_filename</code>	<code>str</code>	<code>read_announcements_archived.txt</code>
<code>scheduled</code>	<code>list</code>	List of scheduled Announcements for today
<code>total_time</code>	<code>int</code>	Total duration of scheduled announcements.
<code>queues</code>	<code>dict</code>	Three separate queues 'High', 'Medium', 'Low'
Methods		
Identifier	Description	
<code>load_announcements() :</code> <code>List(Announcement)</code>	Loads and combines Announcement from <code>deferred_file</code> and <code>announcement_file</code> . Returns a combined list of <code>Announcement</code> objects.	
<code>categorise(announcements)</code>	Sort announcements into queues by priority	
<code>process() :</code> <code>List(announcements)</code>	Schedule announcements until <code>time_limit</code> reached Update <code>scheduled</code> Update <code>total_time</code> Return list of leftover announcements	
<code>display_schedule()</code>	Print scheduled announcements & <code>total_time</code>	
<code>archive_scheduled()</code>	Append to <code>archive_file</code> with separator	
<code>save_deferred()</code>	Save leftover announcements to <code>deferred_file</code> , using <code>output()</code> method in Task 4.2	
<code>clear_today_file()</code>	Empty <code>announcement_file</code> to allow fresh input tomorrow	
<code>run()</code>	Provided – runs full scheduling process You should not modify this method.	

Implement `AnnouncementScheduler` class.

[14]

Task 4.4

Write a Python program that will allow student councillors to input new announcements into `announcement.txt`. The program will request for information on the department, priority, duration and title, and appends this information into `announcement.txt`.

Name your program as

`Task4_4_<name>_<index number>.py`

[2]

[RI/2025/Q4 Cont']

Below shows the result of running the programs on two consecutive days:

Term 1 Day 1

Only announcement.txt file is present - with 7 announcements of various priorities.
No deferred_announcement.txt file is detected by the AnnouncementScheduler program.

announcement.txt

Discipline,High,90,Uniform Violation Rules
Sports,Medium,120,Track Meet Registration
CCA,Low,60,Guitar Club Meeting
Academics,High,150,Exam Briefing
Wellness,Medium,180,Mental Health Talk
Leadership,Low,45,Student Council Promo
Art,Low,90,Art Exhibition Invite



After running Task4_<name>_<index number>.py

Scheduler displays scheduled announcement on **console**

Today's Announcements:
1. Discipline - Uniform Violation Rules (90s)
2. Academics - Exam Briefing (150s)
3. CCA - Guitar Club Meeting (60s)
Total Time: 300s

deferred_announcement.txt created with following contents Empty contents of announcement.txt

Sports,Medium,120,Track Meet Registration
Wellness,Medium,180,Mental Health Talk
Leadership,Low,45,Student Council Promo
Art,Low,90,Art Exhibition Invite

read_announcements_archived.txt is created - with the following contents

Discipline - Uniform Violation Rules (90s)
Academics - Exam Briefing (150s)
CCA - Guitar Club Meeting (60s)
=====

[RI/2025/Q4 Cont ']

At the end of Day 1, the student councillors ran the Task 4.4 program twice to add the following new announcements to the announcement.txt file.

Enter a new announcement :

Department: Principal

Priority (High/Medium/Low): High

Duration (seconds): 120

Title: Welcome message to start of term

Announcement added successfully.

Enter a new announcement:

Department: ICT

Priority (High/Medium/Low): Medium

Duration (seconds): 100

Title: Digital Literacy Week Launch

Announcement added successfully.

Term 1 Day 2

announcement.txt

```
Principal,High,120>Welcome message to start of term
ICT,Medium,100,Digital Literacy Week Launch
```

deferred_announcement.txt

```
Sports,Medium,120,Track Meet Registration
Wellness,Medium,180,Mental Health Talk
Leadership,Low,45,Student Council Promo
Art,Low,90,Art Exhibition Invite
```

After running
Task4_<name>_<index number>.py



Scheduler displays scheduled announcement on console

```
Today's Announcements:
1. Principal - Welcome message to start of term (120s)
2. Sports - Track Meet Registration (120s)
3. Leadership - Student Council Promo (45s)
Total Time: 285s
```

deferred_announcement.txt created with following contents Empty contents of announcement.txt

```
Wellness,Medium,180,Mental Health Talk
ICT,Medium,100,Digital Literacy Week Launch
Art,Low,90,Art Exhibition Invite
```

read_announcements_archived.txt is updated - with the following contents

```
Discipline - Uniform Violation Rules (90s)
Academics - Exam Briefing (150s)
CCA - Guitar Club Meeting (60s)
=====
Principal - Welcome message to start of term (120s)
Sports - Track Meet Registration (120s)
Leadership - Student Council Promo (45s)
```